

Explainable Multimodal AI Framework for Psychiatric Evaluation & Mental-Health Assessment

A deep-learning system that fuses **facial expression**, **speech emotion**, and **behavioural, acoustic & physiological** signals to classify stress severity, estimate Depression / Anxiety / Stress scores, and explain every prediction with clinically-grounded evidence.

Facial · FER-2013 CNN

Speech · Wav2Vec2 SER

Tabular · FT-Transformer

Cross-Modal Attention Fusion

XAI + RAG Decision Support

THEME

Mental-Health Decision Support

TRACK

Multimodal · Explainable AI

TEAM

<Your Team Name>

01 Project Overview

Traditional psychiatric evaluation leans on self-reported questionnaires and clinician interviews. Yet psychological distress leaves a measurable trail across many observable channels — a tightened face, a flattened voice, disturbed sleep, restless typing, an elevated heart rate. Our project builds an **intelligent multimodal assessment framework** that reads these heterogeneous signals together, mirrors how a clinician synthesises evidence, and — crucially — **shows its reasoning** so it can serve as a transparent decision-support tool rather than an opaque black box.

3

MODALITIES FUSED

4

STRESS CLASSES

3

SEVERITY SCORES

34k+

TRAINING SAMPLES

THE ONE-LINE THESIS

A **multimodal, multi-task deep-learning core** makes the predictions; a **RAG-grounded explanation layer** turns those predictions into cited, trustworthy clinical narratives; a **thin agentic orchestrator** sequences the pipeline and answers follow-up questions. The DL model is the brain — retrieval and agents are the voice and hands.

02 Problem Statement & Objectives

Given multimodal inputs — facial images, speech recordings, and 18 numerical behavioural / acoustic / facial / physiological features — the system must automatically assess an individual's mental-health condition, estimate symptom severity, and provide interpretable supporting evidence.

Objective 1 — Status Classification

A multimodal DL framework integrating facial, speech and numerical features to classify each person into **Healthy · Mild · Moderate · Severe Stress**.

Objective 2 — Severity Estimation

A multi-output regression framework to quantitatively estimate **Depression, Anxiety and Stress scores** from the fused indicators.

Objective 3 — Explainable Decision Support

A mechanism that **identifies and quantifies** how each modality (face, speech, behaviour, physiology) contributes to every classification and severity prediction.

Success Criteria

Strong macro-F1 across all four (imbalanced) classes, low RMSE on the three scores, and faithful, human-readable explanations validated against clinical bands.

03 Dataset Understanding

Three complementary sources feed the model. A key engineering insight: the emotion labels in the face and speech sets can be **mapped onto the same 4-level stress scale** as the tabular target, letting all three modalities train toward a shared objective.

Facial · FER-2013
28,709 images · 48×48 grayscale · 7 emotions (Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral). **Heavily imbalanced** — Disgust only 436 vs Happy 7,215.

Speech · RAVDESS
1,440 .wav clips · 24 actors (gender-balanced) · 8 emotions encoded in a 7-part filename. Enables gender-fairness auditing out of the box.

Numerical · Behavioural
4,000 rows · 18 features (sleep, social, typing, blink, MFCC, pitch, heart-rate, HRV, GSR...) · 1 categorical + 3 continuous targets.

Emotion → Stress-level mapping (label unification)

STRESS LEVEL	FACIAL EMOTIONS	SPEECH EMOTIONS
Healthy	Happy, Neutral	Neutral, Calm, Happy
Mild Stress	Sad, Surprise	Sad, Surprised
Moderate Stress	Fear, Disgust	Fearful, Angry
Severe Stress	Angry	Disgust

Note the mapping differs slightly between modalities (Disgust = Moderate for faces but Severe for speech). We treat these as modality-specific priors and let the fusion layer reconcile them, rather than forcing a single hard rule — this is itself an interpretability signal worth surfacing.

The 18 numerical features by domain

BEHAVIOURAL	FACIAL (VIDEO)	ACOUSTIC (AUDIO)	PHYSIOLOGICAL
Sleep Quality · Social Engagement · Daily App Usage · Typing Speed · Session Frequency · Idle Time	Facial-Emotion Variance · Eye-Blink Rate · Smile Intensity · Head-Motion Index	MFCC Mean · MFCC Variance · Pitch Mean · Speech Rate	Heart Rate · HRV Index · Skin Temperature · GSR Level

04 Recommended Approach & Model Choices

This is fundamentally a **supervised multimodal representation-learning problem**, not a retrieval or planning problem. So the core is a deep network with one encoder per modality, an attention-based fusion block, and two task heads sharing that fused representation.

4.1 Per-modality encoders Deep Learning

MODALITY	RECOMMENDED MODEL	WHY THIS ONE	LIGHTWEIGHT FALLBACK
Facial 48x48 grayscale	EfficientNet-B0 CNN (transfer-learned), Grad-CAM hooks on last conv block → 256-d embedding	Best accuracy-per-FLOP for small images; pretrained weights beat training-from-scratch on FER; conv maps give free spatial explanations.	ResNet-18 / a 4-block custom CNN
Speech raw .wav	Wav2Vec2 (self-supervised) fine-tuned for speech-emotion → 256-d embedding	State-of-the-art on speech-emotion; learns directly from waveform, so no hand-crafted feature loss; only 1,440 clips → self-supervised pretraining is a big win.	CNN-BiLSTM over log-Mel spectrograms
Numerical 18 tabular features	FT-Transformer (Feature-Tokenizer + Transformer) → 256-d embedding	Current SOTA deep model for tabular data; attention over features doubles as feature-importance for free, feeding Objective 3.	Residual MLP + BatchNorm

4.2 Fusion — the heart of the system

RECOMMENDED: CROSS-MODAL ATTENTION (INTERMEDIATE FUSION)
Each encoder emits a 256-d token; a small **Transformer fusion block** lets modalities attend to one another so the network learns *per-case* which signal to trust (e.g. down-weight a noisy audio clip). We train with **modality dropout** so the model still works when a modality is missing — essential for real clinical use. Beats naïve early concatenation and late voting, and the attention weights directly answer "which modality drove this?" for Objective 3.

4.3 Multi-task heads

The fused vector feeds two heads trained jointly: a **classification head** (4-way stress class, with Monte-Carlo-dropout uncertainty) and a **multi-output regression head** (Depression, Anxiety, Stress). Because the DASS scores and the stress class describe the same latent state, multi-task learning regularises both and lifts accuracy on each. Loss = class-weighted cross-entropy + uncertainty-weighted MSE (auto-balanced via learned task weights).

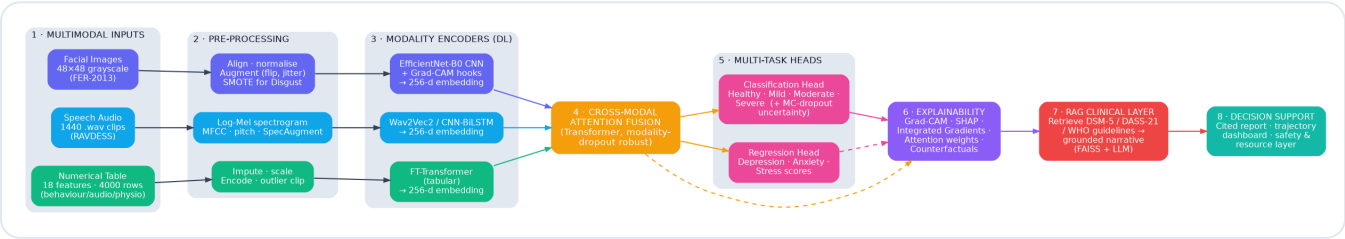
4.4 Agentic AI vs RAG vs "something else" — the honest recommendation

OPTION	RIGHT ROLE HERE?	VERDICT
Agentic AI as the core reasoner	Tempting but wrong: an LLM agent can't reliably read 48×48 faces, waveforms and physiology, and would hallucinate clinical calls.	Not the brain
RAG (retrieval-augmented generation)	Perfect for the <i>explanation</i> layer — after the DL model predicts, retrieve DSM-5 criteria, DASS-21 interpretation bands and WHO guidance to write a grounded, cited narrative with no invented facts.	Use for reporting
Thin agentic orchestration	A lightweight controller sequences preprocess → predict → explain → retrieve → safety-check, and handles interactive follow-ups ("why moderate?", "what changed since last week?").	Use as glue
Multimodal multi-task DL	Does the actual perception and prediction.	The core

WHY THIS WINS OVER "JUST BUILD AN AGENT"

Judges reward correctness *and* trust. A DL core gives measurable accuracy; RAG gives citations a clinician can verify; the agent makes it feel like a product. Making an LLM agent the predictor would sacrifice all three.

05 System Architecture



End-to-end pipeline — three modality encoders → cross-modal attention fusion → multi-task heads → explainability → RAG clinical layer → decision-support output.

STAGE	WHAT HAPPENS
1 • Inputs	Facial images, speech clips, and the 18-feature table enter the pipeline.
2 • Pre-processing	Face alignment + augmentation + SMOTE for the tiny Disgust class; log-Mel + SpecAugment for audio; impute / scale / encode for tabular.
3 • Encoders	EfficientNet-B0, Wav2Vec2/CNN-BiLSTM, FT-Transformer each output a 256-d embedding.
4 • Fusion	Cross-modal Transformer with modality dropout produces one joint representation.
5 • Heads	Classification (4 classes + uncertainty) and multi-output regression (3 scores).
6 • Explainability	Grad-CAM (face), SHAP + attention (tabular), Integrated Gradients (audio), counterfactuals.
7 • RAG layer	FAISS retrieves clinical references; an LLM writes a grounded, cited report.
8 • Decision support	Cited report, longitudinal trajectory dashboard, and a safety/resource layer.

06 Explainability Strategy Objective 3

Each modality gets a faithful, native explanation method; the fusion attention ties them together into a single "contribution budget" the user can read at a glance.

SIGNAL	METHOD	WHAT THE USER SEES
Facial	Grad-CAM / Score-CAM	Heat-map on the face — e.g. tension around the brow & mouth.
Speech	Integrated Gradients on spectrogram	Which time-frequency regions (flat pitch, fast rate) mattered.
Tabular	SHAP + FT-Transformer attention	Ranked feature bars — e.g. low HRV & poor sleep pushed toward Moderate.
Fusion	Attention weights	A modality-contribution meter: 48% physiology · 32% face · 20% speech.
Actionable	Counterfactuals (DiCE)	"If sleep quality rose from 2 → 4, predicted stress drops to Mild."

07 Evaluation Metrics

Classification (Objective 1)

Accuracy · Precision · Recall/Sensitivity · **F1** · **Macro-F1** (equal weight to all four classes — vital given imbalance) · **Weighted-F1** · ROC-AUC · Confusion Matrix.

Regression (Objective 2)

MAE · MSE · **RMSE** (error in original score units) · **R²** · Explained-Variance Score — reported separately for Depression, Anxiety and Stress.

PRIMARY METRIC TO OPTIMISE

Because the classes are imbalanced, **Macro-F1** is our headline classification metric and **RMSE** our headline regression metric — not raw accuracy, which the majority classes would flatter.

08 Out-of-the-Box Features (Winning Edge)

Beyond the required objectives, these differentiators show clinical maturity and product thinking — the things judges remember.

1 Missing-Modality Resilience

Trained with modality dropout, the model degrades gracefully when a webcam, mic, or wearable is unavailable — real clinics rarely have all three. Most teams assume perfect inputs; we don't.

2 Uncertainty-Aware Predictions

Monte-Carlo dropout / evidential heads output a confidence with every call ("Moderate Stress, 78% confidence"). The system knows when to say "refer to a professional" instead of guessing — the hallmark of safe medical AI.

3 Counterfactual "What-If" Coaching

Actionable, non-obvious explanations: *"Raising sleep quality by 2 points would move you from Moderate to Mild."* Turns a diagnosis into a plan.

4 Longitudinal Trajectory Tracking

Re-assessments plot a mental-health trend line over time, so the tool tracks *change* — the metric clinicians actually treat — not just a one-off snapshot.

5 Built-in Fairness & Bias Audit

RAVDESS is gender-balanced, so we ship a fairness dashboard proving the model performs equitably across male / female voices — pre-empting the bias question every health-AI judge asks.

6 RAG-Grounded, Citable Report

Every generated narrative cites DSM-5 / DASS-21 / WHO sources, so no sentence is un-sourced. Trustworthy by construction — a clinician can click through to the evidence.

7 Responsible Crisis-Safety Layer

When severe distress indicators co-occur, the interface calmly surfaces helpline / professional-support resources and frames output as decision-support, never diagnosis. Ethics as a feature, not an afterthought.

8 Privacy-First / On-Device Option

Face and voice never need to leave the device: an on-device inference mode (and a federated-learning training path) directly addresses the biggest barrier to real mental-health adoption.

9 Live Capture Demo

A webcam + mic "assess me now" mode runs the full pipeline live on stage, with the modality-contribution meter animating in real time — an unforgettable demo moment.

09 Complete Tech Stack

LAYER	TOOLS
Core language	Python 3.11
Deep learning	PyTorch · PyTorch Lightning · torchvision · torchaudio
Vision encoder	timm (EfficientNet-B0 / ResNet-18) · OpenCV · albumentations
Speech encoder	HuggingFace Transformers (Wav2Vec2) · librosa · torchaudio (SpecAugment)
Tabular encoder	pytorch-tabular / rtdl (FT-Transformer) · scikit-learn · imbalanced-learn (SMOTE)
Explainability	Captum (Grad-CAM, Integrated Gradients) · SHAP · DiCE (counterfactuals)
RAG layer	LangChain / LlamaIndex · FAISS or Chroma · sentence-transformers · an LLM (open-source Llama-3 or hosted API)
Experiment tracking	Weights & Biases or MLflow · Hydra (configs)
Backend / serving	FastAPI · Uvicorn · ONNX Runtime (fast inference)
Frontend	React + Vite · Tailwind CSS · Recharts / D3 (dashboards) · WebRTC (live capture)
Data & infra	pandas · NumPy · DVC (data versioning) · Docker · docker-compose

10 Project Working Directory

```
hack4health-mindscope/
├── data/                                # raw + processed (DVC-tracked, git-ignored)
│   ├── raw/                            # facial/ speech/ numerical.csv
│   └── processed/                      # spectrograms, aligned faces, scaled tables
├── configs/                           # Hydra YAML: model, data, train, fusion
├── notebooks/                         # EDA, label-mapping, error analysis
├── src/
│   ├── data/                          # datasets, loaders, augmentation, SMOTE
│   ├── encoders/                     # face_cnn.py speech_w2v.py tabular_ft.py
│   ├── fusion/                       # cross_modal_attention.py, modality_dropout.py
│   ├── heads/                        # classifier.py regressor.py uncertainty.py
│   ├── models/                       # multimodal_net.py (assembles the whole graph)
│   ├── explain/                      # gradcam.py shap_tab.py ig_audio.py counterfactual.py
│   ├── rag/                          # index_builder.py retriever.py report_generator.py
│   ├── agent/                        # orchestrator.py (pipeline + follow-up Q&A)
│   ├── train/                        # train_multitask.py losses.py metrics.py
│   └── utils/                        # seed, logging, fairness_audit.py
├── api/                               # FastAPI app: /predict /explain /report /assess-live
├── frontend/                          # React + Tailwind multipage UI
│   └── src/pages/                     # Dashboard, Assessment, Explain, Trajectory, Report
├── models_store/                     # checkpoints + ONNX exports
├── knowledge_base/                   # DSM-5 / DASS-21 / WHO docs for RAG
├── tests/                            # unit + integration tests
├── docker/                           # Dockerfile, docker-compose.yml
├── requirements.txt
├── dvc.yaml
└── README.md
```

11 Work Frame & Execution Timeline

A phase-based plan sized for a hackathon sprint — front-load a working end-to-end skeleton, then deepen.

- **Phase 0 — Setup & EDA** [Hours 0–4](#)
Repo scaffold, environment, download data, exploratory analysis, confirm the emotion → stress label mapping, quantify class imbalance.
- **Phase 1 — Unimodal baselines** [Hours 4–12](#)
Train each encoder alone (face CNN, speech Wav2Vec2, tabular FT-Transformer). Lock preprocessing. These baselines are also the ablation study judges love.
- **Phase 2 — Fusion + multi-task core** [Hours 12–22](#)
Assemble cross-modal attention fusion, add both heads, train jointly with modality dropout & uncertainty. Hit target Macro-F1 / RMSE.
- **Phase 3 — Explainability** [Hours 22–30](#)
Wire Grad-CAM, SHAP, Integrated Gradients, attention meter and counterfactuals; validate they're faithful and readable.
- **Phase 4 — RAG + agent + safety** [Hours 30–38](#)
Build the clinical knowledge index, grounded report generator, thin orchestrator, and the crisis-safety / resource layer.
- **Phase 5 — UI + live demo** [Hours 38–46](#)
Connect FastAPI to the React multipage UI, wire the live-capture demo and trajectory dashboard, run the fairness audit.
- **Phase 6 — Polish & pitch** [Hours 46–48](#)
Metrics tables, ablation results, rehearse the story, record a backup demo video.

RESPONSIBLE-AI NOTE (STATE THIS ON STAGE)

This is a **decision-support** tool to assist professionals, **not** a diagnostic device. It surfaces evidence and uncertainty, keeps sensitive face/voice data private, and points people toward qualified human support — the framing that turns a clever model into a responsible health product.